WebClient Computing
jQuery Event

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Modifying DOM Elements

- jQuery provides methods to modify
  - Change attribute values
  - Element contents (text)
  - Add and remove classes

Adding or Removing Classes

- In real programming, if you want to change style what are you going to do?
- Method 1
  - An element has a class assigned, namely “classic”
  - If you want to change how the element looks like, change the style of the given element one by one
- Method 2
  - An element has a class assigned, namely “classic”
  - If you want to change how the element looks like, create another class called “modern” and change the element’s class to modern

How to Add or Remove Class From Element

- addClass(class_name)
  - Add a class attribute to the given element
- removeClass(class_name)
  - From a given element, remove the class attribute

Add or Remove Class Example

- jquery_class.html and jquery_class.js

```html
<!DOCTYPE html>
<html>
<head>
  <title>jQuery exercise</title>
  <script type="text/javascript" src="http://code.jquery.com/jquery-2.2.3.min.js"></script>
  <style>
    .red {color:red;}
    .blue {color:blue;}
    .larger {font-size:3em;}
  </style>
</head>
<body>
  <div id="target">Demo Element</div>
</body>
</html>
```

```javascript
$('#target').addClass('blue');
$('#target').addClass('larger');
$('#target').removeClass('blue');
$('#target').addClass('red');
$('#target').removeClass('red larger');
~
```

Quick Notes

- We can use combinator in CSS
  - Remember CSS Selector Combinator?
- Direct parent-child
  - $(‘div > a’)
- Descendent
  - $(‘div a’)
- Adjacent sibling
  - $(‘div + a’)
- General sibling
  - $(‘div ~ a’)

Handling Event with jQuery

- jQuery event handling
  - Remove event from HTML
  - Easy to update
  - Can be dynamic
- Access the event sender
  - Use this keyword
  - this object is DOM element (not jQuery object)
- $().
  - click
  - keypressed
  - mouseout

Event Handling With jQuery

- jQuery provides methods for registering event handlers
- Event handler registration using CSS selector, class, tag names, …
- Type of Events
  - click() – mouse click event
  - dblclick() – mouse double click event
  - mouseenter() – when a mouse pointer enters an element
  - mouseleave() – when a mouse pointer leaves an element
  - keypress() – when a keyboard is pressed
  - Full list - http://api.jquery.com/category/events/

Recap – Event Object

- Every event incudes an event object
  - EX: pageX and pageY for muse coordinates
  - EX: keyCode for a button pressed that triggered the event
  - Access the object using a keyword event

```
<!DOCTYPE html>
<html>
<head>
  <title>Javascript event</title>
  <script type="text/javascript" src="event_keyboard.js"></script>
</head>
<body>
<input type="text" size="40" onkeypress="key_pressed(event)">
<p id="key-demo"></p>
</body>
</html>
```

Handling Click Event With jQuery

- $(SELECTOR).click(function(){});
- When a mouse is clicked to the given selector, the given event handler is performed

```
$('#target-id').click(function (event) {
  output = "User clicked on " + event.pageX + "/" + event.pageY;
  $('#display').text(output);
})

$('.target-class').click(function() {
  output = "Class is clicked";
  $('#display').text(output);
})

$('div').click(function() {
      $('#display').text("div tag is clicked");
})
```

```
<body>
  <div id="target-id">ID - Click me</div>
  <div class="target-class">Class - Click me<div>
  <p id="display"></p>
  <script type="text/javascript" src="jquery_event.js"></script>
</body>
```

jquery_mouse_event.html/js

Compare Click Event Javascript - jQuery

- Javascript

```html
<!DOCTYPE html>
<html>
<head>
  <title>Javascript event</title>
  <script type="text/javascript" src="event.js"></script>
</head>
<body>

<p>Click the button to display the date.</p>

<button onclick="displayDate()">The time is?</button>

<p id="demo"></p>

</body>
</html>
```

```javascript
function displayDate() {
    document.getElementById("demo").innerHTML = Date();
}
```

jQuery

```javascript
$('#target-id').click(function (event) {
  output = "User clicked on " + event.pageX + "/" + event.pageY;
  $('#display').text(output);
})

$('.target-class').click(function() {
  output = "Class is clicked";
  $('#display').text(output);
})

$('div').click(function() {
      $('#display').text("div tag is clicked");
})
```

```html
<body>
  <div id="target-id">ID - Click me</div>
  <div class="target-class">Class - Click me<div>
  <p id="display"></p>
  <script type="text/javascript" src="jquery_event.js"></script>
</body>
```

Using this keyword in Javascript

- Represent an object that can access the object itself
- In a function, this keyword means the function itself
- In an event handler, this keyword represents an object that triggers an event

jQuery focus and blur Event

- focus() event
  - Fires when a input field gets focus (when mouse pointer is in it)
- blur() event
  - Fires when a input field looses focus

```
$('input').focus(function() {
  $(this).next('span').removeAttr('hidden');
});

$('input').blur(function() {
  $(this).next('span').attr('hidden', 'hidden');
});
```

```
<body>
  <div>
    NAME: <input type="text"></input><span hidden="hidden">Type your name</span>
  </div>
  <div>
    E-MAIL: <input type="text"></input><span hidden="hidden">Type your e-mail address</span>
  </div>
  <script type="text/javascript" src="jquery_form.js"></script>
</body>
```

jquery_form.js/html

Triggering Event Within Event Handler

- In an event handler, calling trigger invokes an event
  - $(SELECTOR).trigger(EVENT_NAME)
  - EX: $(‘button’).trigger(‘click’)

```html
<body>
  <button id="button-1">Click me - Button 1</button>
  <button id="button-2">Click me - Button 2</button>
  <p id="clicked-ID"></p>
  <p id="display"></p>
  <script type="text/javascript" src="jquery_trigger_event.js"></script>
</body>
```

```javascript
$('#button-1').click(function() {
  $('#display').text("Button 1 is clicked");
  $('#clicked-ID').text($(this).attr('id'));
});

$('#button-2').click(function() {
  $('#button-1').click()
  $('#clicked-ID').text($(this).attr('id'));
});
```

jquery_trigger_event.html js

Reference

- edx jQuery course
- w3schools.com
- http://api.jquery.com/